

Chien-Wen (Wayne) Pan

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SUMMARY

Postdoctoral Associate working on an upper-limb robot for rehab and proprioception studies from zero to demo for a university spinout company. During the doctoral study, my work utilizes Koopman-inspired methods and physics-informed machine learning, bridging biomechanical and physiological signals with human intentions for wearable robots and healthcare applications. Experienced in modeling and characterizing of electromechanical actuators, traditional and modern control algorithms and simulations for robots, and multi-modal sensory. Passionate about developing sensory intelligence and adaptive robotics systems.

SKILLS

Domain Robotics simulation, Reinforcement learning, Nonlinear modeling, System identification, Signal processing, Physics-informed machine learning, Mechatronics

Programming and Tools: Python, PyTorch, TensorFlow, MuJoCo, C/C++, MATLAB/Simulink, Git/Github, Linux, ROS2

Hardware and Prototyping: Embedded systems and edge compute (Raspberry Pi, Jetson nano, Arduino, STM32), 3D printing, laser cutting, Solidworks, LabVIEW

EDUCATION

Arizona State University

Ph.D., Systems Engineering

Tempe, AZ

Dec. 2025

Dissertation: Model-based Machine Learning for Physiological Wearables in Healthcare and Human-Robot Interactions

Committees: Sangram Redkar, Thomas Sugar, Kevin Nicols

University of Michigan

M.S., Robotics

Ann Arbor, MI

Apr. 2018

National Cheng-Kung University

B.S., Aeronautics and Astronautics

Taiwan

Jun. 2014

PUBLICATIONS

- C.W. Pan, M. Rahmani, and S. Redkar, 2026, "EMG-based gait estimation and prediction using Koopman-inspired method," *IEEE Transactions on Biomedical Engineering*.
- C.W. Pan, M. Rahmani, and S. Redkar, 2026, "Integrating biophysical dynamics with data-driven method for blood pressure (BP) estimation using photoplethysmography (PPG)," *Biomedical Signal Processing and Control*.
- C.W. Pan, S. Redkar, 2025, "Physical Fatigue Evaluation Using Muscle Modeling," *Proc. of 2025 IEEE International Conference on Consumer Electronics (ICCE)*.
- J.R. Yu, C.W. Pan, K.Y. Tsai, et al., 2025, "Assessment Device for the Stability of Osseo-integration," *Proc. of 2025 IEEE International Conference on Consumer Electronics (ICCE)*.
- J.Y. Hsu, C.W. Pan, Sh.W. Chung, et al., 2024, "Automatically Correcting and Compensating Measurement Data of Wearable Inertial Sensors for Gait Analysis," *Proc. of SPIE*.
- U.H. Lee, C.W. Pan, and E.J. Rouse, 2019, "Empirical characterization of a high-performance exterior-rotor type brushless DC motor and drive," *Proc. of 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.
- P.T. Pan, C.W. Pan, A.B. Wang, et al. 2016, "A precision capillary coating system and applications," *Smart Science*.
- H.T. Li, C.W. Pan, H.I. Chi, et al., 2015, "Inertial Sensing Based Assessment Methods to Quantify the Effectiveness of Post Stroke Rehabilitation," *Sensors*.

EXPERIENCE

University of Minnesota Twin Cities (Human Sensorimotor Control Lab)

Postdoctoral Associate

Minneapolis, MN

Mar. 2026 – Present

- Co-authored a \$1.25M NSF SBIR Phase 2 grant proposal based on successful hardware and software prototype validation.
- Integrated BLDC actuators and sensors with STM32 boards, implementing CAN bus communication between parent and child nodes for an upper-limb robot.

- Designed closed-loop control algorithms and integrated the hardware with Unity-based gamified UI.
- Conducted 100+ comprehensive customer discovery interviews with stroke survivors, clinical directors, and physical therapists for NSF I-Corps summer 2026 cohort.

Arizona State University (Robotics and Dynamical Systems Group)

Tempe, AZ

Research Assistant / Ph.D. Candidate

Aug. 2021 – Nov. 2025

- Developed a two-stage Koopman-based pipeline predicting lower-limb kinematics from EMG signals, achieving 95% accuracy and stable near-future multi-step prediction.
- Trained a PIML model for BP estimation using PPG signals, integrating cardio-dynamics with nonlinear lifting; validated on 120 subjects in ICU condition.
- Engineered consumer-facing TinyML solution deployed on Arduino Nano33 for exercise fatigue estimation from physiological and kinematic signals.

Arizona State University (IdeaLAB)

Tempe, AZ

Research Assistant

Aug. 2020 – Jul. 2021

- Developed Python models of human lower limbs, performing inverse and forward dynamics to estimate joint torques for biomechanical analysis.
- Prototyped a hybrid origami-based device featuring layer jamming integration, achieving tunable structural stiffness for passive actuation systems.

University of Michigan (Neurobionics Lab)

Ann Arbor, MI

Graduate Independent Research Study

Sep. 2017 – Jun. 2018

- Designed and built a power test bench to characterize a high-performance, exterior-rotor BLDC motor and drive system for an open source bionic leg.

National Central University (Diagnostics and Biomedical Measurement Lab)

Taiwan

Undergrad Research Intern

Jun. 2014 – Aug. 2014

- Applied SVM regression to IMU data for objective post-stroke rehabilitation assessment, enhancing clinical evaluation.

INDEPENDENT/EXTERNAL COOPERATING ONGOING RESEARCH

Independent Research

Model-Predictive Control in MyoSuite simulation

Jan. 2026 – Present

- Expanding doctoral EMG-based state estimation framework to incorporate model-predictive control within the MuJoCo simulation environment.

External Cooperate

Motor Neuroscience

May. 2026 – Present

- EEG data analysis in MATLAB (EEGLAB/ERPLAB) and conduct statistical evaluations to process raw EEG data during ipsilateral positioning tasks.

External Cooperate

Diagnostics and Biomedical Measurement

Jan. 2024 – Present

- Osseointegration assessment techniques and devices.
- Wearable inertial sensors for biomechanical analysis.

TEACHING EXPERIENCE

MAE 318 System Dynamics and Control, Arizona State University

Teaching Associate

Jan. 2020 – May 2020

- Led an undergrad Modeling and Control lab for 46 students, providing technical guidance and evaluating lab reports.

AWARDS

- ASU Manufacturing Systems and Networks (MSN) Achievement Award

Jan. 2023